CO2 - AT1

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Q1) House-Price Prediction Using Linear Regression

Problem Understanding

Predict house price using:

- X_1= Area in sq. ft.
- X_2= Number of bedrooms
- Y= Price in ₹ lakhs

Method Selection

Use Multiple Linear Regression:

$$\hat{Y} = b_0 + b_1X_1 + b_2X_2$$

The least-squares method estimates the coefficients.

Solution

The estimated coefficients are:

$$b_0 = 8.5047,\ b_1 = 0.04252,\ b_2 = -0.28037$$

Therefore:

$$\boxed{\hat{Y} = 8.5047 + 0.04252X_1 - 0.28037X_2}$$

Interpretation

- Intercept 8.5047: estimated base price is ₹8.50 lakhs.
- Area coefficient 0.04252: each additional square foot increases price by approximately ₹4,252, keeping bedrooms constant.
- Bedroom coefficient -0.28037: one additional bedroom reduces the predicted price by ₹0.28 lakhs when area is fixed.

The negative bedroom coefficient occurs because the dataset is very small and area and bedrooms are strongly related.

Q2) Linear Classification of Spam Email

Problem Understanding

Classify an email using:

- X_1= frequency of “Offer”
- X_2= frequency of “Win”
- Output: 1 = Spam, 0 = Not Spam

New email:

$$X_1 = 2,\ X_2 = 1$$

Method Selection

Use a linear classification score:

$$g(X) = b_0 + b_1X_1 + b_2X_2$$

Classify as spam when $g(X) \geq 0.5$.

Solution

The fitted model is:

$$g(X) = -0.10 + 0.35X_1 + 0.05X_2$$

Decision boundary:

$$-0.10 + 0.35X_1 + 0.05X_2 = 0.5$$
$$\boxed{7X_1 + X_2 = 12}$$

For $X_1 = 2, X_2 = 1$:

$$g(X) = -0.10 + 0.35(2) + 0.05(1)$$
$$g(X) = 0.65$$

Since $0.65 \geq 0.5$:

$$\boxed{\text{Spam}}$$

Interpretation

The new email lies on the spam side of the decision boundary. Its frequent use of the words "Offer" and "Win" increases its spam score.

Q3) Disease Prediction Using Bayes' Theorem

Problem Understanding

Find the probability of disease when:

$$Fever = 1,\ Headache = 0$$

There are three disease cases and two non-disease cases.

Method Selection

Use Naive Bayes:

$$P(D' \mid FH) = \frac{P(D)P(F \mid D)P(H \mid D)}{\sum_C P\,(C)P(F \mid C)P(H \mid C)}$$

Solution

Prior probabilities:

$$P(Disease) = \frac{3}{5} = 0.6$$
$$P(No\ Disease) = \frac{2}{5} = 0.4$$

Conditional probabilities:

$$P(Fever = 1 \mid Disease) = \frac{3}{3} = 1$$
$$P(Headache = 0 \mid Disease) = \frac{1}{3}$$
$$P(Fever = 1 \mid No\ Disease) = \frac{0}{2} = 0$$

Disease score:

$$0.6 \times 1 \times \frac{1}{3} = 0.2$$

No-disease score:

$$0.4 \times 0 \times \frac{1}{2} = 0$$

Therefore:

$$P(D\text{،}isease \mid Fever = 1Headache = 0) = \frac{0.2}{0.2 + 0}$$
$$\boxed{P(Disease) = 1.0 = 100\%}$$

Interpretation

Based on this small dataset, a patient with fever and no headache is predicted to have the disease. The 100% result occurs because no non-disease patient has fever.

Q4) Linear Discriminant Classification

Problem Understanding

Treating the two input columns as $X_1$and X_2:

- Class C_1: $(2\text{،}3), (3\text{،}4)$
- Class C_2: $(5\text{،}6), (6\text{،}7), (4\text{،}5)$

Classify the new point $(3\text{،}3)$.

Method Selection

Use nearest-mean linear discriminant analysis with equal class priors and equal identity covariance.

$$g_i(X) = \mu_i^T X - \frac{1}{2}\mu_i^T \mu_i$$

Solution

Class means:

$$\mu_1 = (2.5\text{،}3.5)$$
$$\mu_2 = (5\text{،}6)$$

Discriminant functions:

$$g_1(X) = 2.5X_1 + 3.5X_2 - 9.25$$
$$g_2(X) = 5X_1 + 6X_2 - 30.5$$

Decision boundary:

$$g_1(X) = g_2(X)$$
$$2.5X_1 + 2.5X_2 - 21.25 = 0$$
$$\boxed{X_1 + X_2 = 8.5}$$

For (3, 3):

$$g_1 = 8.75,\ g_2 = 2.5$$

Therefore:

$$\boxed{(3, 3)\ belongs\ to\ C_1}$$

Interpretation

The new point is closer to the centre of C_1, so the discriminant model assigns it to class C_1.

Q5) Bayesian Logistic Regression for Customer Churn

Problem Understanding

Predict customer churn using:

- Monthly charges
- Tenure in months
- Output: 1 = Churn, 0 = No Churn

New customer:

$$Charges = 650,\ Tenure = 9$$

Method Selection

Use Bayesian Logistic Regression:

$$P(Y = 1 \mid X) = \frac{1}{1 + e^{-z}}$$
$$z = \beta_0 + \beta_1 X_1 + \beta_2 X_2$$

Since the question does not specify a prior, assume standardized features and a standard normal prior for the coefficients.

Solution

After standardization, the approximate MAP model is:

$$z = 0.2803 + 0.6607Z_{Charges} - 0.7246Z_{Tenure}$$

For charges 650and tenure 9:

$$Z_{Charges} = \frac{650 - 620}{172.05} = 0.1744$$
$$Z_{Tenure} = \frac{9 - 10}{3.4059} = -0.2936$$
$$z = 0.2803 + 0.6607(0.1744) - 0.7246(-0.2936)$$
$$z \approx 0.608$$
$$P(Churn) = \frac{1}{1 + e^{-0.608}}$$
$$\boxed{P(Churn) \approx 64.75\%}$$

Since the probability is above 50%:

$$\boxed{\text{Customer is predicted to churn}}$$

Interpretation

Higher monthly charges increase churn probability, while longer tenure reduces it. The new customer has a moderate-to-high churn risk.

Q6) Customer Classification Using KNN

Problem Understanding

Classify the customer $(A, ge = 32Income = 50)$using $k = 3$.

Method Selection

Use Euclidean distance:

$$d = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$

Solution

Distance from $C_1 = (25, 40)$, Class A:

$$d = \sqrt{(32 - 25)^2 + (50 - 40)^2} = \sqrt{149} = 12.21$$

Distance from $C_2 = (30, 60)$, Class A:

$$d = \sqrt{104} = 10.20$$

Distance from $C_3 = (35, 80)$, Class B:

$$d = \sqrt{909} = 30.15$$

Distance from $C_4 = (40, 20)$, Class B:

$$d = \sqrt{964} = 31.05$$

Three nearest neighbours:

1. C_2– A
2. C_1– A
3. C_3– B

Majority class:

$$\boxed{A}$$

Interpretation

Two of the three nearest customers belong to Class A. Therefore, the new customer is classified as Class A.

Q7) Entropy and Information Gain

Problem Understanding

Calculate the entropy of Play and Information Gain for Outlook.

The target contains:

- Yes = 2
- No = 2

Method Selection

Entropy:

$$H(S) = -\sum P(c)\log_2 P(c)$$

Information Gain:

$$IG(S, A) = H(S) - H(S \mid A)$$

Solution

Target entropy:

$$H(Play) = -\frac{2}{4}\log_2\frac{2}{4} - \frac{2}{4}\log_2\frac{2}{4}$$
$$\boxed{H(Play) = 1\ bit}$$

For Outlook:

- Sunny: 2 No, entropy = 0
- Overcast: 1 Yes, entropy = 0
- Rain: 1 Yes, entropy = 0

Conditional entropy:

$$H(Play \mid Outlook) = 0$$

Information Gain:

$$IG(Outlook) = 1 - 0$$
$$\boxed{IG(Outlook) = 1\ bit}$$

Root node:

$$\boxed{Outlook}$$

Interpretation

Outlook completely separates the available Yes and No examples. Therefore, it is the best root node for the decision tree.

Q8) Linear Separating Hyperplane

Problem Understanding

The data points are:

- $+1$: $(2, 2), (4, 4)$

- -1: $(2,4), (4,2)$

The positive and negative points occupy opposite corners of a square.

Method Selection

Check whether a single straight line can separate both classes.

Solution

The pattern is an XOR arrangement:

(2,4) -1 (4,4) +1

(2,2) +1 (4,2) -1

Any straight line that separates one positive point from a negative point will incorrectly classify another point.

Therefore:

$$\boxed{\text{No linear separating hyperplane exists}}$$

A nonlinear classification rule is:

$$(x_1 - 3)(x_2 - 3) > 0 \Rightarrow +1$$
$$(x_1 - 3)(x_2 - 3) < 0 \Rightarrow -1$$

The nonlinear decision boundaries are:

$$\boxed{x_1 = 3 \text{ and } x_2 = 3}$$

Interpretation

The dataset is not linearly separable. A nonlinear SVM with a polynomial or RBF kernel is required.

Q9) Comparison of Classification Models

Problem Understanding

Calculate accuracy, precision and recall for Logistic Regression and Naive Bayes.

Method Selection

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$
$$Recall = \frac{TP}{TP + FN}$$

Logistic Regression

$$TP = 40,\ FN = 10,\ FP = 5,\ TN = 45$$

Accuracy:

$$\frac{40 + 45}{100} = 0.85$$
$$\boxed{Accuracy = 85\%}$$

Precision:

$$\frac{40}{40 + 5} = 0.8889$$
$$\boxed{Precision = 88.89\%}$$

Recall:

$$\frac{40}{40 + 10} = 0.80$$
$$\boxed{Recall = 80\%}$$

Naive Bayes

$$TP = 35,\ FN = 15,\ FP = 10,\ TN = 40$$

Accuracy:

$$\frac{35 + 40}{100} = 0.75$$
$$\boxed{Accuracy = 75\%}$$

Precision:

$$\frac{35}{35 + 10} = 0.7778$$
$$\boxed{Precision = 77.78\%}$$

Recall:

$$\frac{35}{35 + 15} = 0.70$$
$$\boxed{Recall = 70\%}$$

Interpretation

Logistic Regression performs better in all three metrics:

Model	**Accuracy**	**Precision**	**Recall**
Logistic Regression	85%	88.89%	80%
Naive Bayes	75%	77.78%	70%

Therefore:

$$\boxed{\text{Logistic Regression is the better model}}$$

Q10) One Iteration of K-Means

Problem Understanding

Data points:

$$(2, 4), (3, 5), (8, 9), (9, 10)$$

Initial centroids:

$$C_1 = (2, 4),\ C_2 = (8, 9)$$

Method Selection

Assign each point to its nearest centroid using Euclidean distance, then calculate the new cluster means.

Solution

Point	**Distance to C_1**	**Distance to C_2**	**Cluster**
(2,4)	0	7.81	C_1
(3,5)	1.41	6.40	C_1

Point	Distance to C_1	Distance to C_2	Cluster
(8,9)	7.81	0	C_2
(9,10)	9.22	1.41	C_2

Clusters:

$$Cluster\ 1 = \{(2,4),(3,5)\}$$
$$Cluster\ 2 = \{(8,9),(9,10)\}$$

New centroid of Cluster 1:

$$C_1' = \left(\frac{2+3}{2},\frac{4+5}{2}\right)$$
$$\boxed{C_1' = (2.5,4.5)}$$

New centroid of Cluster 2:

$$C_2' = \left(\frac{8+9}{2},\frac{9+10}{2}\right)$$
$$\boxed{C_2' = (8.5,9.5)}$$

Interpretation

After one iteration, the two nearby lower-valued points form Cluster 1, while the two higher-valued points form Cluster 2. The updated centroids represent the centre of each cluster.